Assignment 1

PCCST501–ComputerNetworks

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How the Internet Delivers a Web Page

1. TCP/IP Protocol Stack and Role of Each Layer

TCP/IP organises internet communication into five layers, each handling a specific job and depending on the layer below it.

ACaseStudyonYouTube(www.youtube.com)

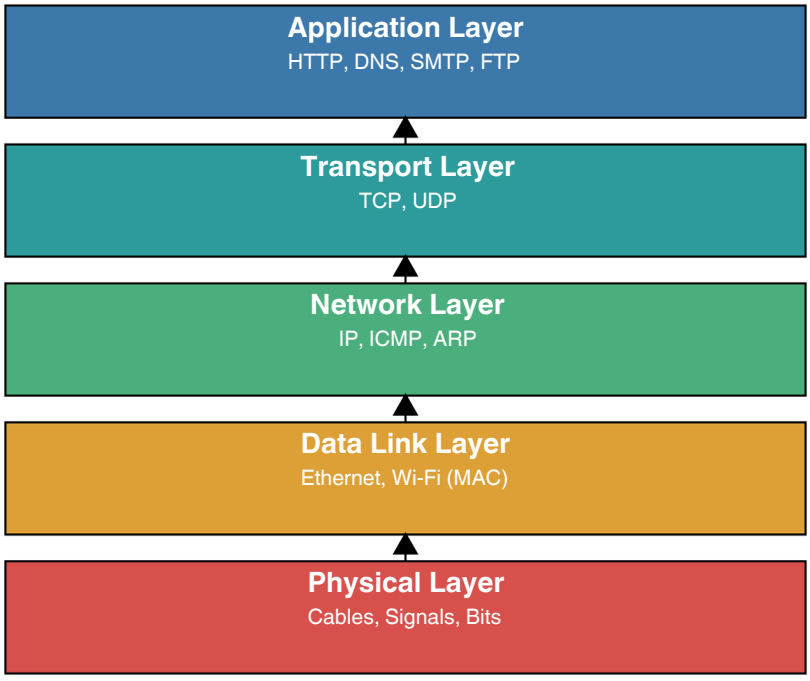


Fig 1: Five-layer TCP/IP stack used to load the YouTube homepage

- **Application Layer –** closest to the user. HTTP fetches the page, DNS resolves the domain name.
- **Transport Layer –** TCP provides reliable, ordered, connection-oriented delivery between browser and server.
- **Network Layer –** IP addresses each packet and routes it across networks toward the YouTube server.
- **Data Link Layer –** wraps packets into frames and moves them across one physical link (LAN/Wi-Fi) using MAC addresses.

- Physical Layer- sends the frame as raw bits — electrical, optical, or radio signals — over the medium.

2. Role of DNS

DNS converts a domain name like www.youtube.com, which humans can remember, into the numeric IP address that computers actually use for routing.

- Browser checks its local/OS cache first for a saved IP address.
- If not cached, the request goes to a DNS resolver (usually the ISP's).
- Resolver queries a root server → .com TLD server → YouTube's authoritative server.
- The IP address returned is cached briefly (TTL) to speed up future visits.

3. How HTTP Requests and Receives the Webpage

HTTP works on a request–response model over a TCP connection (port 80, or 443 for HTTPS).

- **Request:** browser sends a GET with the request line (method, URL, version) and headers like Host and Accept-Language.
- **Response:** server replies with a status line (e.g. 200 OK), headers (content type, length), and the body — HTML, CSS, JS, images.
- **Multiple requests:** a page like YouTube's homepage has many embedded objects, so the browser sends several GETs, usually reusing one persistent HTTP/1.1 connection.

4. How TCP Provides Reliable Communication for HTTP

TCP is connection-oriented and reliable, which is why HTTP is built on top of it. A three-way handshake (SYN, SYN-ACK, ACK) sets up the connection before any HTTP data moves.

- **Error control –** sequence numbers and acknowledgements detect lost/corrupted segments and trigger retransmission.
- **Ordered delivery –** sequence numbers let the receiver reassemble segments in the correct order.
- **Flow control –** matches sender's rate to the receiver's buffer capacity.
- **Congestion control –** adjusts sending rate based on network conditions.

5. Client–Server or Peer-to-Peer Architecture?

YouTube follows a **Client–Server** architecture, not P2P.

- YouTube's servers/CDN are always-on and centrally hosted by Google.
- The browser is purely a client — it requests content and closes once it's received.
- All videos, metadata and comments are served centrally, not shared peer-to-peer like BitTorrent.
- Client and server roles are fixed; a browser can never act as the server.

6. HTTP vs FTP

Basis	HTTP	FTP
Purpose	Requests and delivers web pages and embedded objects	Transfers files between client and server
Connections	One TCP connection, persistent by default (HTTP/1.1)	Two connections — control (port 21) and data (port 20)
State	Stateless; sessions faked using cookies	Stateful — tracks session and current directory
Port	80 (HTTP) / 443 (HTTPS)	21 (control) / 20 (data)
Data handling	Optimised for small, frequent request/response exchanges	Optimised for large file transfer (ASCII/binary modes)

7. How Cookies Improve User Experience

HTTP is stateless, so cookies let a server remember a client across requests: the server sends a Set-Cookie header once, and the browser attaches it to every later request to that domain.

Example (YouTube): after logging in and watching a few videos, a cookie stores the login state and viewing preferences. On the next visit YouTube keeps the user logged in, remembers language/region, resumes watch-history-based recommendations, and keeps autoplay/quality settings — without asking again.

8. Complete Communication Flow

From URL entry to page render — DNS resolves the address, TCP sets up a reliable connection, HTTP carries the request/response, and the browser renders the result:

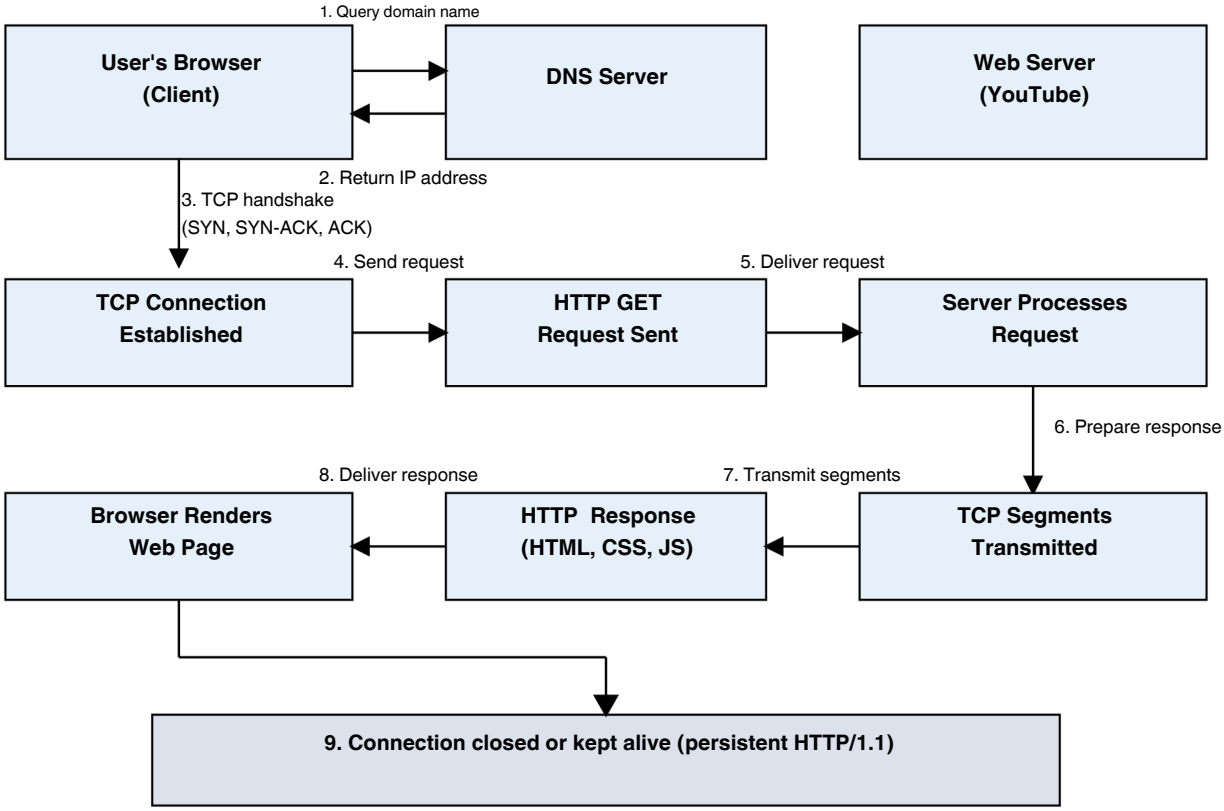


Fig 2: End-to-end communication flow between browser and YouTube server

9. Conclusion

A single click on YouTube sets off a tightly coordinated sequence: DNS resolves the name, TCP guarantees reliable delivery, and HTTP carries the actual request and response — all riding on the layered TCP/IP architecture. Without these application-layer protocols working together, everyday tasks like browsing, streaming, emailing or transferring files would not be possible. Cookies add to this by making sessions feel continuous and personalised. Studying this flow shows how the internet hides enormous technical complexity behind a simple, near-instant experience that billions of users rely on every day.